
PROGRAM OVERVIEW

Building real systems for real challenges

Kibler AI Solutions Corp. is a registered Ohio C-corporation (EIN: 33-3532580, State ID: 5358264) founded in February 2025 in Salem, Ohio. The company develops AI infrastructure for community resilience, with active engagement from Youngstown State University's APEX Accelerator, SBDC, and the Youngstown Advanced Manufacturing Innovation Center (YAMIC).

This summer, the company's research contributors — including **Damen Kibler (18)** and **Aaron Kibler (17)** — are working on active federal challenge submissions under the direct mentorship of founder **Bradford Kibler Jr.** The work spans signal processing, biological systems modeling, hardware prototyping, and AI governance architecture.

Bradford Kibler Jr.

BK

Founder & CEO, Lead Composer-Architect. 25 years across skilled nursing, union carpentry, injection molding, and structural steel. Self-taught systems architect.

Damen Kibler

DK

Research Contributor — RATML state machine development, signal processing testing, system validation

Aaron Kibler

AK

Research Contributor — SIEVPS architecture, system testing, hardware integration support

Three federal challenges, one architecture

All three tracks share a common technical foundation — the **Guardian architecture** — a constitutional AI framework built around the RATML state machine, Helios governance layer, and MyceliumNet mesh routing. Each track applies this foundation to a different domain.



NASA Space to Soil Challenge

Phase 1 Submission — Deadline: May 4, 2026

MyceliumNet Orbital Edge — Adaptive onboard processing for SmallSat Earth observation supporting regenerative agriculture. Features RATML-driven spectral band selection, lunar-cadenced sensing, Doppler-phosphorescence sensor fusion, and a physical ground-truth validation system using a custom-built biochar gasifier and mycelium colonization protocols.

Python

Signal Processing

Earth Observation

Biochar Production

Mycology

SmallSat Architecture



ACL Caregiver AI Prize Challenge

Intent to Apply: April 15, 2026 — Phase 1: July 31, 2026

Guardian Caregiver Intelligence Platform — AI-enabled monitoring and decision-support for family caregivers and direct care workers. Features ambient health monitoring through RATML state detection, Philips WiZ LED color-mapped alerts, one-touch iPhone Shortcut activation, and a constitutional governance layer ensuring the AI never overrides human caregiver authority.

Healthcare AI

IoT Integration

Caregiver Support

Privacy Architecture

Accessibility



Community Mesh Broadband

In development with APEX Accelerator & YAMIC

MyceliumNet Community Mesh — A rural broadband model adapted for small-town terrain, with working PoE spine MVPs pulling 1.1 Gbps backbone and ~765 Mbps client speeds at 0% packet loss. Designed for digital equity in underserved communities like Salem, Ohio. Architecture review underway with YSU engineering faculty.

Networking

PoE Infrastructure

Digital Equity

Rural Broadband

Hardware Deployment

LEARNING OUTCOMES

Skills developed through this program

Participants gain hands-on experience across a range of STEM disciplines through direct contribution to active federal research submissions — not simulations or classroom exercises.

AI & Machine Learning

State machine design, signal processing, decision systems

Software Development

Python, JavaScript, system architecture, version control

Biological Systems

Mycology, biochar production, regenerative agriculture

Hardware Engineering

Gasifier fabrication, network infrastructure, IoT

Technical Writing

Federal proposals, whitepapers, documentation

Business Operations

Corporate structure, federal registration, challenge compliance

Data Science

Sensor fusion, spectral analysis, signal classification

Ethics & Governance

AI safety, constitutional architecture, privacy design

For Educators

This program involves direct contribution to federally-recognized innovation challenges (NASA, HHS/ACL) through a registered U.S. corporation with active university partnerships. The work is mentored, documented, and produces verifiable deliverables.

Potential academic alignment includes independent study credit, Career & Technical Education (CTE), project-based learning portfolios, dual enrollment or early college STEM credit, and National Honor Society service/leadership documentation. We welcome conversation with guidance counselors, CTE coordinators, or STEM department heads about structuring credit recognition.

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University & federal partnerships

Kibler AI Solutions Corp. operates within an active network of institutional support spanning federal innovation programs, university research centers, and state-level business development infrastructure.

APEX Accelerator
Ohio APEX at YSU
Gov't contracting

SBDC
YSU Small Business
Dev. Center

YAMIC
Advanced Mfg.
Innovation Center

Valley Partners
Regional economic
development

NASA
Space to Soil
Challenge

HHS / ACL
Caregiver AI
Prize Challenge

VERIFY & LEARN MORE

Official challenge links

All federal challenge information is publicly verifiable through the links below.

NASA Space to Soil Challenge
Official challenge page — [nasa.gov](#)

ACL Caregiver AI Prize Challenge
Official challenge page — [acl.gov](#)

Ohio APEX Accelerator at YSU
Federal contracting assistance — [apex-ysu.org](#)

YSU College of STEM
Youngstown State University — [ysu.edu](#)

